

Coupling-gated cortical coherence: what a field theory of conscious unity must assume, and what it predicts

Sebastian Bobadilla-Suarez
`sebastian.bobadilla.s@gmail.com`

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Abstract

A field account of cortical coordination holds that spatially distributed neural processes are coupled in part through the endogenous extracellular electromagnetic field. We ask what such an account must assume to reach a globally consistent, self-referential state, and what it predicts before it gets there. Its physical core is a noisy mean-field phase model whose coupling drifts through a synchronization threshold. If recovery of coherence after sleep is gated by that crossing, it is not free to take an arbitrary time course. For as long as the phase distribution tracks the stationary branch, it must show a square-root onset and a von Mises concentration–coherence relation with no fitted shape parameter. The tracking condition is part of the prediction, and the simulations measure its cost: ramp speed sets the apparent onset exponent, so an exponent fitted to a driven recovery trace reports the drive as much as the mechanism. A coupling plasticity descending a squared-drift objective preserves coherence while moving away from the environmental structure it should learn, below a shuffled-label comparison: dissipation minimization does not imply representational learning. A Lean 4 formalization composes finite capacity, Landauer’s

bound, the noisy Kuramoto transition, sheaf gluing and Banach’s theorem into one conditional theorem, isolating the eight hypotheses it consumes; a common toy model satisfies all eight. We defend as a philosophical commitment, not a theorem, the identification of the glued section with the unity of a conscious episode and of its reflexive fixed point with a minimal self. Testing the account requires independent calibration of effective cortical coupling and an awakening experiment separating slow coupling change from alternative recovery mechanisms.

1 Introduction

Neural activity changes across sleep, awakening and anesthesia, but a change in synchrony does not by itself identify the physical interaction responsible for it. A field account makes a stronger proposal: spatially distributed neural processes are coordinated partly through the endogenous extracellular electromagnetic field, and changes in the medium can change that coordination (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Whether that mechanism is sufficient to support the collective state such accounts require remains an empirical question.

Such an account has three requirements, and they order what follows. Coupling must be strong enough to sustain coherent collective activity; the local descriptions that activity carries must be compatible enough to glue into a single one; and that glued state must support an accurate model of itself. The first requirement is physical and testable with current methods, and most of this article is about testing it. The second and third are where

17 the mathematics is, and where the assumptions accumulate.

18 The first requirement admits a definite claim. If cortical coupling is slowly
19 gated—if recovery of coherence after sleep is driven by a coupling parameter
20 drifting through a synchronization threshold—then the recovery is not free
21 to take an arbitrary time course. For a recovery slow enough that the phase
22 distribution tracks the stationary branch, the theory fixes a square-root onset
23 and a one-parameter relation between phase concentration and coherence,
24 with no fitted shape parameter. That is a constraint an experiment can
25 meet or violate. How slow is slow enough is itself part of the claim, and
26 our simulations bound it from the other side, by showing which parts of the
27 constraint survive finite populations and finite ramp speeds and which do
28 not.

29 The contribution has four parts. The first is the coupling-gated recovery
30 hypothesis itself: a stationary coherent branch with square-root onset, a von
31 Mises concentration–coherence relation, and an experimental protocol with
32 failure conditions declared in advance (Sections 2 and 3). The second is a
33 constraint on how such data may be analyzed. The apparent onset expo-
34 nent measured from a finite ramp is a function of ramp speed across the
35 sweep of Section 4, so an exponent fitted to a recovery trace measures the
36 ramp at least as much as the mechanism—and that holds for any analysis
37 reading a critical exponent off a driven transition, not only for ours. The
38 third is a negative result about plasticity: a coupling update descending
39 a squared-drift objective preserves coherence while moving *away* from the

environmental structure it is expected to learn, below a shuffled-label comparison. Dissipation minimization does not imply representational learning. The fourth is a machine-checked composition (Section 5) in which finite information capacity, Landauer’s bound, a predictive dissipation bound, the noisy Kuramoto transition, sheaf gluing and Banach’s fixed-point theorem yield one conditional theorem, with the eight hypotheses that composition consumes isolated and named; a common toy model satisfies all eight, so their conjunction is not empty.

A prose table of assumptions would list the same eight items, so the fourth contribution needs a word about method. What the Lean development adds is that it makes the list non-negotiable. The characteristic failure in an argument of this shape is not a false step but a silent change of subject: a theorem about a stationary density read as a claim about a trajectory, an inequality about a register read as a claim about learning, agreement of phases read as compatibility of representations. Type-checking forbids each of those substitutions, and an assumption needed to make a step go through appears as a hypothesis in the term rather than as a word in a sentence. Formalization does not make the assumptions true; it makes them impossible to lose.

The physical model and the formal composition belong in one article because they are two halves of one commitment. The coupling-gated prediction is worth testing under this framework, rather than as one more synchronization model, only because the coherent state it produces is claimed to do

63 further work: to support a globally consistent representation and a reflexive
64 fixed point that we identify with phenomenal unity and a minimal self.

65 The distinction that does that work is between phases agreeing and de-
66 scriptions fitting together, and it is concrete. Take two cortical regions whose
67 receptive territories overlap, each carrying a probability distribution over
68 what occupies the shared part. Their oscillators can be locked to the same
69 phase at every instant while one region's distribution on the overlap places an
70 object left of fixation and the other places it right. The phases agree; the de-
71 scriptions do not, and no single distribution over the shared territory restricts
72 to both. A theory that identifies unity with synchrony cannot mark that dif-
73 ference, and so cannot separate a brain whose regions oscillate together from
74 one whose regions are about the same world. The sheaf condition marks
75 exactly it: local states glue to one global section when their restrictions to
76 shared regions coincide, and otherwise not at all. Coherence enters as the
77 physical condition under which compatibility can be maintained, not as a
78 substitute for it, and the self-prediction map then asks the further ques-
79 tion of whether the glued state contains an accurate model of itself. That
80 identification is a commitment, not a theorem, and Section 6 states it, ar-
81 gues for it, and says what would count against it. Its separation from the
82 physical tests is what makes both assessable: the cortical coupling hypothe-
83 sis and the recovery prediction can fail while the mathematical composition
84 stands, and the composition can stand while the identification is wrong. The
85 established ingredients (Landauer, 1961; Still et al., 2012; Sakaguchi, 1988;

86 Banach, 1922) therefore appear here for their biological composition, not for
 87 any new fundamental physical law.

88 **2 A coupling-gated model of cortical coher-** 89 **ence**

90 **2.1 Dynamics and stationary solutions**

91 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

92 Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and
 93 K_{ij} has inverse-time units, with angles treated as dimensionless. The uni-
 94 form mean-field convention is $K_{ij} = K/N$. Frequency heterogeneity and
 95 phase diffusion are separate parameters; a frequency-spread width cannot be
 96 substituted for D when applying the identical-frequency stationary theory.

97 For identical frequencies in their rotating frame, the assumed stationary
 98 density is

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}. \quad (2)$$

99 The I_n are modified Bessel functions, and ψ is the collective phase. Lean

100 verifies normalization, the density's circular order parameter, and the self-
 101 consistency equation. At $K \leq K_c = 2D$, the only nonnegative self-consistent
 102 order parameter is zero. At $K > K_c$, exactly one positive solution coex-
 103 ists with zero; the positive branch is continuous at threshold and strictly
 104 increasing with K .

105 The near-threshold expansion gives

$$\frac{r^2}{K - K_c} \rightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

106 This is a result about the stationary solution branch. Of the coherent branch
 107 we prove existence, uniqueness, continuous emergence at threshold and strict
 108 growth in K , but not that it is dynamically selected. The dynamical mean-
 109 field limit—propagation of chaos—is the missing link between the finite sys-
 110 tem and the stationary description (Sznitman, 1991), and no theorem in this
 111 development closes it. Exchange symmetry alone does not justify factoriza-
 112 tion of the joint law, particularly for an ensemble with an unpinned collective
 113 phase.

114 A spatial version replaces the finite sum by an integral against a kernel
 115 $K(x, y)$. We normalize the substrate measure to total mass one, so the mean-
 116 field strength is a kernel average, not a quantity that grows with the arbitrary
 117 measure of the substrate. Reduction of a heterogeneous spatial kernel to a
 118 scalar K is an additional modelling step, and the threshold in Eq. (2) does
 119 not automatically apply to every spatial or signed network.

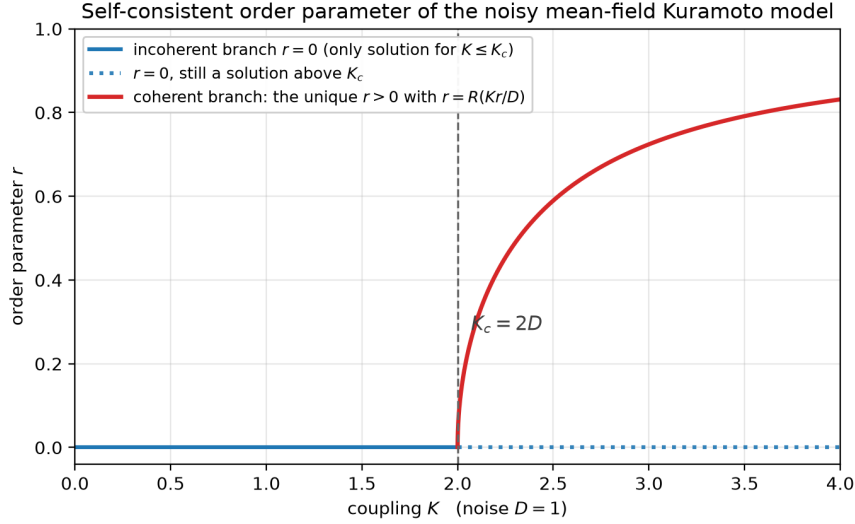


Figure 1: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the same integral expressions used in the formalization; it illustrates the static theorems and does not establish dynamical selection.

2.2 Physical identification and calibration

Regular mesh refinement gives convergence of a discrete coupling-energy quadrature to its continuum integral: scalar energy convergence, not convergence of trajectories and not a construction of a physical kernel. The model posits a continuum kernel and separately identifies its cortical realization with endogenous electromagnetic coupling. The formal predicate requires a jointly continuous kernel, a normalized substrate, a mean-field strength equal to its spatial average, and no single site providing more than half of that mean-field coupling. Toy witnesses satisfy these conditions; none is a

129 model calibrated to cortex.

130 Extracellular fields can modulate spike timing (Fröhlich & McCormick,
131 2010; Anastassiou & Koch, 2015), and field-based models have been proposed
132 for integrating distributed neural information (Pinotsis & Miller, 2022; Mc-
133 Fadden, 2020). These observations motivate an effective interaction without
134 establishing its magnitude relative to phase diffusion, its spatial averaging
135 rule, or its causal role in experience; changes in phase organization under
136 propofol (Bastos et al., 2021) likewise do not measure a crossing of the model’s
137 coupling threshold.

138 The physical-unit conversion is the sharpest open limitation of the frame-
139 work. An estimate proportional to $N\Delta tf$, using population size, a spike-time
140 shift and oscillation frequency, is dimensionless. It cannot be compared di-
141 rectly with an inverse-time diffusion parameter. A conversion such as

$$K_{\text{eff}} = \gamma N s E f \tag{4}$$

142 requires a rate calibration γ , where s is the time shift per unit field and E the
143 field amplitude. The saved field-strength calculations are conditional on this
144 calibration and on the chosen spatial aggregation; they do not independently
145 establish $K_{\text{eff}} > 2D$ in cortex. Determining the interaction kernel and phase
146 diffusion from the same physical system is necessary before the threshold
147 becomes a quantitative biological claim; Section 3.2 makes that measurement
148 the first item of the proposed experiment.

149 3 Recovery after awakening: a conditional 150 prediction

151 Sleep-associated changes in extracellular space and clearance motivate a pos-
152 sible slow control of coupling (Xie et al., 2013). We hypothesize that recovery
153 after awakening includes a change in effective coupling $K(t)$, mediated by
154 extracellular geometry, on a slower timescale than phase dynamics. Neither
155 the monotonic relation between geometry and coupling nor its role in hu-
156 man sleep inertia follows from the cited physiological observation; the model
157 requires both to be measured.

158 If $K(t)$ rises through K_c and the phase distribution tracks the coherent
159 stationary branch, Eq. (3) predicts

$$r(t) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad t > t_0, \quad \dot{K}(t_0) > 0. \quad (5)$$

160 The ideal stationary description has a subthreshold interval followed by con-
161 tinuous onset. Saturation requires a specified subsequent coupling trajectory;
162 monotonic coupling alone does not determine a logistic or any unique sig-
163 moid. The onset time is fixed by $K(t_0) = K_c$ only if $K(t)$ is independently
164 known. When coupling is inferred from the same coherence trace, this is not
165 a parameter-free prediction of latency.

166 Critical slowing down limits the adiabatic prediction of Eq. (5) near
167 threshold, and a finite population carries nonzero fluctuating order below

168 it. An observable onset therefore requires a predeclared threshold above the
 169 finite-size floor, and a finite-rate trajectory need not show an infinite-slope
 170 cusp. Section 4.1 quantifies the limitation.

171 **3.1 Distributional and temporal tests**

172 The stationary model makes a second prediction: phase concentration and
 173 coherence obey $r = I_1(a)/I_0(a)$, with no fitted shape parameter. Its appli-
 174 cability depends on the stationary-density assumption and the observation
 175 model, and the relation is common to the whole von Mises family, so agree-
 176 ment with it identifies neither ephaptic coupling nor this framework against
 177 alternative synchronization models. Within the same ansatz $a/r = K/D$ for
 178 $r > 0$, so the limiting threshold ratio follows from the density relation instead
 179 of testing it, and ratios near the finite-size floor are especially unstable.

180 The estimators must be independent, which is the constraint most easily
 181 violated in practice. Estimating a by inverting the Bessel ratio from r would
 182 enforce the relation being tested, and linear statistics on wrapped phases
 183 estimate neither quantity.

184 **3.2 A protocol for the awakening experiment**

185 The requirements above define a design rather than a single measurement,
 186 and the design is where the content of the hypothesis lives. An adequate test
 187 needs all six of the following.

- 188 **1. Coupling and diffusion in the same units.** Estimate K and D
189 from the same cortical regime, in inverse-time units, separating phase
190 diffusion from quenched frequency spread and from observation noise.
191 Without this the threshold $K > 2D$ is not an evaluable statement
192 about cortex.
- 193 **2. Geometry calibrated to coupling.** A diffusion-MRI observable or
194 an astrocytic signal is an indicator of extracellular geometry, not a mea-
195 surement of K . Its mapping to effective coupling must be established
196 separately, with its own uncertainty.
- 197 **3. Phase estimation at adequate spatial resolution.** At least 100
198 spatial samples per estimate. Estimate r by the circular resultant
199 length and a by an unweighted regression of log-density on the co-
200 sine of phase relative to the circular mean. With fewer sites, pooling
201 over time changes the estimand to a spatiotemporal distribution and
202 must be stated as such.
- 203 **4. An independent recovery endpoint.** Behavioral recovery measured
204 separately from the phase trace, so that latency is not defined by the
205 signal whose latency is at issue.
- 206 **5. A predeclared onset criterion.** A coherence threshold fixed above
207 the finite-size floor before unblinding, chosen in the knowledge that a
208 finite ramp smears the cusp.

209 **6. Model comparison under matched observation.** Coupling-gated
 210 recovery compared against exponential, flexible sigmoidal and other
 211 slow-recovery models under identical temporal filtering and observation
 212 noise. Latency alone cannot distinguish the mechanism, because many
 213 processes create delays.

214 The design fails informatively in three ways: failure of the stationary
 215 phase-density relation over a sufficiently broad concentration range; absence
 216 of the proposed coupling–geometry association; or a recovery trace better
 217 explained by an alternative after finite-rate and observation effects are ac-
 218 counted for. Two apparent failures are not failures. A linear apparent onset
 219 is evidence against the adiabatic approximation only where that approxima-
 220 tion is independently justified, and it does not falsify the field hypothesis
 221 under arbitrary ramp speeds. The absence of a fold in the stationary equa-
 222 tion excludes stationary coexistence of two positive branches in this model,
 223 but does not exclude finite-rate hysteresis or adjudicate theories of ignition.

224 **4 Numerical controls**

225 The simulations characterize the finite-size, finite-rate regime that the sta-
 226 tionary theory leaves open, and they bear on whether the prediction of Sec-
 227 tion 3 is measurable at all. None of them discharges a hypothesis of the
 228 composition in Section 5. Each control uses explicit seeds and compact saved
 229 summaries, and the numerical values below are generated from those arti-

230 facts. The supplement gives parameters, estimators, failure criteria and ad-
 231 ditional panels. Identical-frequency controls use phase diffusion rather than
 232 a quenched frequency-spread surrogate, and order plots display the $1/\sqrt{N}$
 233 reference.

234 4.1 Ramp speed sets the apparent onset exponent

235 The dynamic-ramp study (Fig. 2) uses $N = 2000$, $D = 1$ and 32 replicas at
 236 four coupling speeds. Escape is the first of three consecutive saved samples at
 237 $r \geq 0.2$ after threshold. Only 5/32 replicas escape within the fastest window.
 238 The three slower ramps have mean coupling delays 0.1947, 0.0684 and 0.0253,
 239 with a fitted delay exponent 0.443. The apparent onset exponent changes
 240 from 0.964 at the fastest speed to 0.417 at the slowest.

241 The result cuts against the prediction of Section 3 as much as against
 242 any other reading of a driven transition. The exponent recovered by fitting
 243 such a transition is a joint function of the mechanism and the drive rate. Re-
 244 porting an onset exponent from a recovery trace therefore says little about
 245 the underlying transition unless the ramp rate is independently known and
 246 the finite-size floor is declared; the stationary square-root law cannot be im-
 247 posed on an arbitrary recovery trace, and its absence is not by itself evidence
 248 against a coupling-gated mechanism. The specified deviation-based critical
 249 ramp speed is not bracketed by the sweep.

250 The dynamical-selection control shows escape toward a coherent branch
 251 above threshold, but retains quantitative finite-size discrepancies. At $K =$

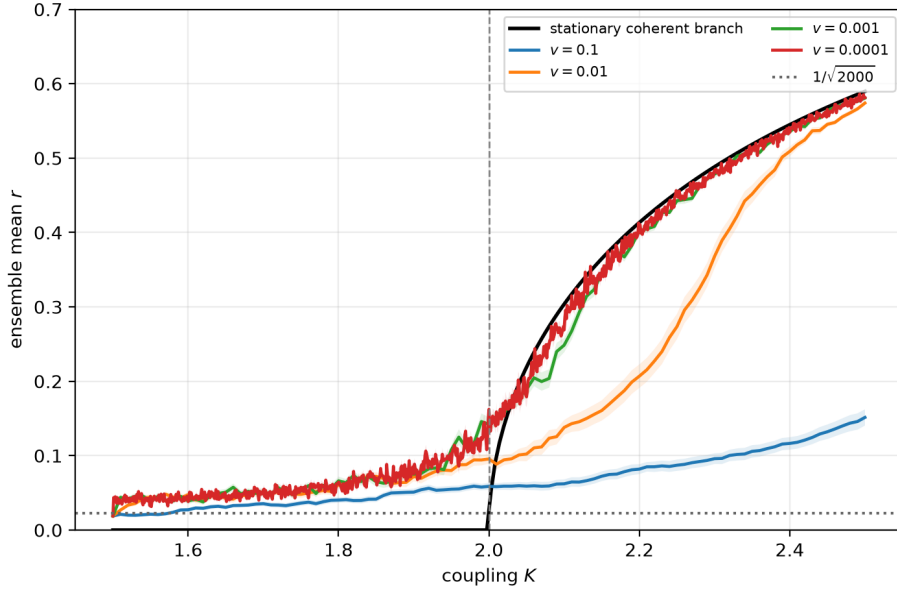


Figure 2: Finite-rate departures from the stationary branch. Ensemble-mean order is plotted against the increasing coupling for four ramp speeds v , with bands of one standard error across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape; the stationary branch is not a universal time course for recovery.

252 2.8, steady mean order is 0.67735 against the static value 0.68270. The
 253 fitted growth-rate slope is 0.43240, compared with the theoretical slope $1/2$.
 254 The propagation-of-chaos diagnostics show decreasing subcritical dependence
 255 with increasing population size. In the supercritical unpinned ensemble,
 256 however, pair correlation remains substantial because realizations share a
 257 random collective orientation. This negative unconditional result does not
 258 settle a conditional mean-field limit after removal of that orientation.

259 4.2 Spatial coupling, frustration and calibration

260 A periodic two-dimensional sheet with row-normalized exponential coupling
261 remains coherent in the sampled empirical decay-length band under the de-
262 clared total coupling and frequency spread. At decay lengths 0.1, 0.2 and
263 0.3 mm, steady order is 0.9424, 0.9427 and 0.9427. This is one finite param-
264 eter regime; fixing total coupling while changing range tests fragmentation,
265 not whether cortex supplies that total coupling.

266 A row-balanced Dale-law network shows that row balance alone does not
267 guarantee a floor-level incoherent baseline. In a separately specified weak-
268 synapse regime, a uniform modulatory term produces an operational coher-
269 ence crossing at K_{eff}/D between 1.8994 and 1.9387. That ratio is dimension-
270 less; carrying it over to cortex needs the rate calibration γ of Section 2.2,
271 which neither this sweep nor the spatial control supplies.

272 4.3 Squared-drift plasticity does not learn structure

273 A coupled system that descends a dissipation-like objective is widely expected
274 to come, in doing so, to mirror the structure of its input; the predictive
275 dissipation bound of Section 5 invites that reading. This control tests the
276 expectation directly in the present model, and the expectation fails.

277 For fixed phases, consider the squared-drift objective

$$\sigma(K) = \frac{1}{D} \int \left[\omega(x) + \int K(x, y) \sin(\theta_y - \theta_x) d\mu(y) \right]^2 d\mu(x). \quad (6)$$

278 It is a quadratic functional of an affine image of the kernel. The formalization
 279 constructs the bounded drift operator A and its adjoint on the relevant L^2
 280 spaces, obtaining $\nabla\sigma = (2/D)A^*(AK + \omega)$, and the specified gradient flow
 281 decreases this objective at fixed phases. The objective is not generally a
 282 thermodynamic entropy-production rate, which requires probability-current
 283 information. Whether biological couplings follow this flow is an additional
 284 modelling question.

285 The joint control evolves noisy phases with clustered frequencies and
 286 nonzero grand-mean drive, while updating symmetric couplings every 50 fast
 287 steps. Couplings remain nonnegative, hollow and normalized to a fixed to-
 288 tal. The symmetric-edge derivative of $\frac{1}{2}\sum_i v_i^2$ is $(v_i - v_j)\sin(\theta_j - \theta_i)$, so both
 289 endpoints contribute. Resource constraints and evolving phases remove the
 290 fixed-phase monotonicity guarantee.

291 Across three seeds, second-half order lies between 0.9811 and 0.9815, and
 292 matched frozen-coupling controls maintain similar coherence. The distance
 293 to the true cluster template increases in every adaptive run, with worse
 294 changes than the shuffled-label comparison (Fig. 3). The template is a co-
 295 variance of shared latent cluster signals, not an input to the update or a
 296 measured covariance of the constant frequencies.

297 The objective descends, coherence is maintained, and the coupling matrix
 298 moves away from the generating structure—below the level of a random
 299 relabelling. Coherence is therefore not a proxy for representational alignment
 300 in this regime, and a plateau in the objective establishes neither stationarity

301 of the adaptive process nor a thermodynamic entropy-production rate.

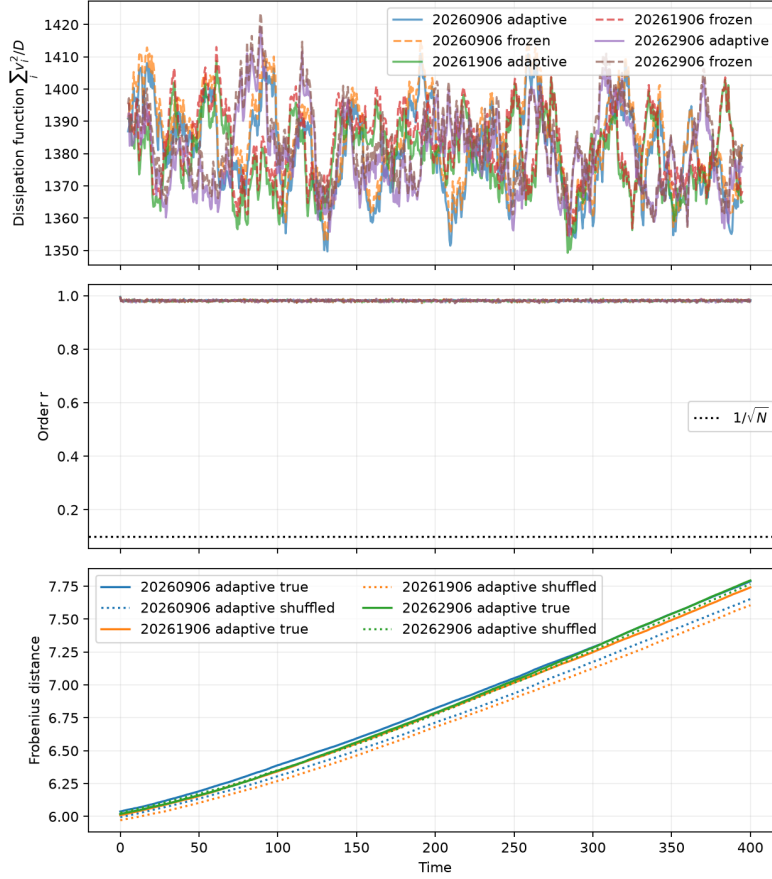


Figure 3: Joint fast/slow dynamics with matched frozen-coupling controls. Top: smoothed squared-drift dissipation, with frozen controls dashed. Middle: order and the finite-size reference. Bottom: distances to true cluster templates (solid) and node-label shuffles (dotted). Coherence persists, but structural distances increase. The result limits the proposed connection between dissipation minimization and environmental structure.

302 4.4 The estimation protocol demonstrated on public 303 EEG

304 Item 3 of Section 3.2 specifies estimators whose behavior on real recordings
305 is not obvious, so we exercise them on public propofol-sedation EEG (Bajwa
306 et al., 2025). The purpose is to show what the protocol produces and where
307 it is fragile, not to test the awakening hypothesis, for which these data are
308 the wrong design.

309 Two fragilities appear immediately. The implemented estimator pools 62
310 channels over a 100-ms window, so it estimates a spatiotemporal distribu-
311 tion rather than an instantaneous spatial one—a departure from the protocol
312 forced by channel count, and one that any scalp implementation will face.
313 Bipolar and common-reference montages give different residuals, making ref-
314 erencing a material analysis choice rather than a preprocessing detail.

315 The bipolar results are compatible with the Bessel relation over the ob-
316 served concentration range. That range is too narrow to distinguish the
317 curve from its linear approximation at the observed scatter, so the check is
318 uninformative about the relation itself. An informative test needs a broader
319 concentration range and explicit treatment of source mixing, finite samples
320 and temporal pooling; intracranial or high-density recordings across a gen-
321 uine state transition are the natural setting.

322 5 Thermodynamic constraints and the for- 323 mal composition

324 The synchronization model of Section 2 is one link of a longer conditional
325 construction, whose nine nodes are:

326 **n1** a finite phase space has bounded information capacity; **n2** a
327 broken symmetry forces a boundary, where the field leaves its vac-
328 uum manifold; **n3** the boundary erases, and erasure costs heat;
329 **n4** that dissipation bounds nonpredictive memory; **n5** discrete
330 couplings coarse-grain to a continuum limit; **n6** a specified phys-
331 ical field realizes that limiting coupling; **n7** above threshold the
332 field carries a unique coherent order parameter; **n8** compatible
333 local states glue to a unique global section; **n9** that section is the
334 unique fixed point of a self-prediction map.

335 Node-level results are established or cited; what the composition consumes
336 are the eight implications between them. Figure 4 shows where they sit and
337 what kind of assumption each one is, and Table 1 states them, classified by
338 their role in the composition rather than by their plausibility. In particular,
339 finite capacity does not imply a symmetry-breaking boundary: **n1** and **n2** are
340 independent roots, even though the composition can accept an implication
341 between them as an argument.

342 5.1 Capacity, boundaries and predictive memory

343 For a finite state space X , Shannon entropy satisfies $H(p) \leq \log |X|$, with
344 equality only for the uniform distribution. A stream with more possible

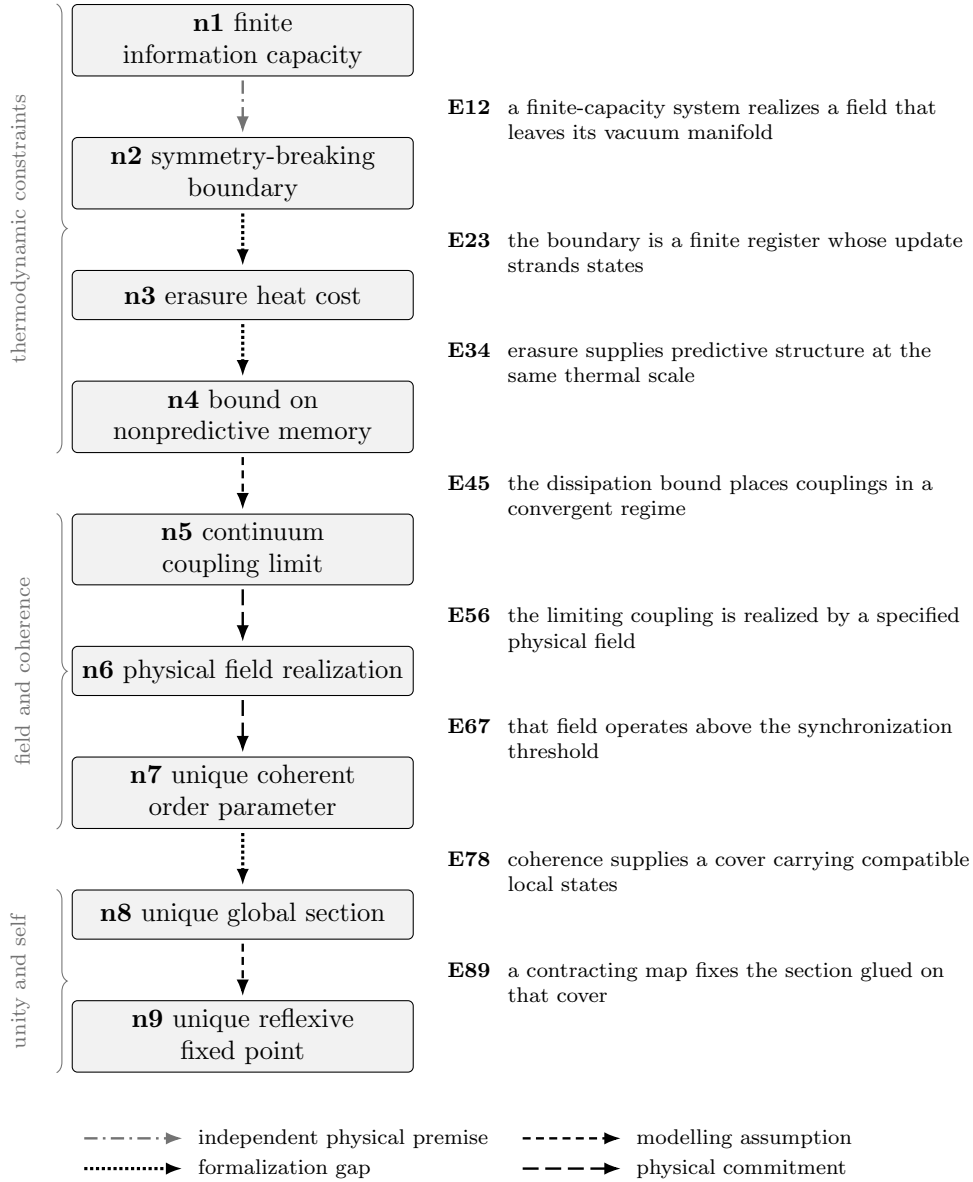


Figure 4: The conditional composition. Each node is an established or cited result; each arrow is an implication the composition consumes and no theorem here supplies, styled by the kind of assumption it carries. The three formalization gaps bracket the two physical commitments, so the chain’s empirical content sits between links of a different kind on either side. E12 is drawn as a premise because n1 and n2 are independent roots. Table 1 states each connection in full.

Edge	Status	Required connection
E12 capacity $\rightarrow$ boundary	Independent physical premise	A finite-capacity system realizes a field that leaves its vacuum manifold. Capacity alone does not imply this.
E23 boundary $\rightarrow$ erasure	Formalization gap	The boundary is associated with a finite register whose update leaves states unreachable.
E34 erasure $\rightarrow$ prediction	Formalization gap	Erasures supplies a predictive structure with the same thermal scale and heat budget, and nonzero wasted memory.
E45 prediction $\rightarrow$ continuum	Modelling assumption	A predictive dissipation bound places coupling energies in a convergent coarse-graining regime.
E56 continuum $\rightarrow$ field	Physical commitment	The limiting coupling is realized by a specified field satisfying the kernel conditions. Its identification with cortex is empirical.
E67 field $\rightarrow$ coherence	Physical commitment	That field operates above the synchronization threshold.
E78 coherence $\rightarrow$ unity	Formalization gap	Coherence supplies a cover with coupling at least the named mean-field value, reached by relaxation and carrying compatible local states.
E89 unity $\rightarrow$ self	Modelling assumption	The state glued on that same cover is fixed by a contracting map mediated by its local self-encoding.

Table 1: The assumptions consumed by the conditional composition theorem, each an implication between two of the nine nodes above. Node-level results do not establish these connecting implications. The complete theorem map and witnesses are in Supplemental Material, Table S1.

345 histories than $|X|$ cannot be encoded injectively in that register. Separately,
 346 a continuous field on a connected substrate that reaches distinct components
 347 of a disconnected vacuum manifold must leave that manifold somewhere,
 348 the simplest topological-defect construction (Kibble, 1976). Relating this
 349 locus to a particular finite information-processing register is the additional
 350 hypothesis E12 carries.

351 A non-surjective update on a finite register is non-injective. Under the
 352 specified system–bath model and thermal assumptions, erasure carries a Lan-
 353 dauer heat cost (Landauer, 1961). Finite capacity does not imply that every
 354 update erases, and logical many-to-one processing must be distinguished from
 355 its physical implementation (Norton, 2005; Shenker, 2000). The formaliza-
 356 tion records the physical heat identity as an obligation of each model and
 357 supplies an explicit one-bit eraser satisfying it.

358 For a system state X_t driven by an environmental signal S_t , the predictive
 359 dissipation bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (7)$$

360 Under the no-back-action Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$, data process-
 361 ing makes the bracket nonnegative (Still et al., 2012). The bound constrains
 362 nonpredictive memory at a given dissipation budget. It does not specify
 363 a learning rule, maximize prediction, or require the internal representation
 364 to reproduce environmental geometry—the point the plasticity control of

365 Section 4 makes concretely. Connecting this budget to Landauer heat ad-
 366 ditionally requires nonpredictive information to be bounded by the entropy
 367 erased by the register. The formal bridge derives Eq. (7) from Landauer’s
 368 bound under that identification; it does not derive the identification.

369 A bound $D_{KL} \leq \Delta t \sigma$ also cannot force $D_{KL} \rightarrow 0$ unless its right-hand
 370 side tends to zero, and Gibbs’ inequality proves nonnegativity of relative en-
 371 tropy, not a general identification of relative entropy with physical entropy
 372 production. These distinctions prevent an informational inequality from be-
 373 ing read as a derivation of environmental learning, Bayesian inference, or
 374 semantic content.

375 5.2 Compatible states and reflexive fixed points

376 A sheaf formalizes local descriptions and their restriction to smaller regions.
 377 If local sections agree on every overlap, the sheaf condition gives a unique
 378 global section. In the probability-sheaf construction, this is consistency of
 379 local density information, and it is not implied by agreement of oscillator
 380 phases: the development includes a counterexample with equal phases but
 381 incompatible densities on an overlap. Section 6 rests on this distinction,
 382 and the model excludes such configurations only through its compatibility
 383 hypothesis.

384 A thermodynamic cover couples this gluing construction to a phase con-
 385 figuration at a potential minimum. For the deterministic positive-coupling
 386 model, convergence to that minimum is proved under an explicit initial basin

387 condition; other equilibria prevent an unconditional conclusion. The station-
388 ary noisy order parameter, the deterministic relaxation, and the cover must
389 therefore be connected by E78 rather than identified without a hypothesis.
390 Neither field continuity nor synchronization alone guarantees a globally com-
391 patible representation.

392 A reflexive extension equips global sections with a complete metric and a
393 predictive map formed by restricting a state to a local encoding region and
394 reading that encoding out. If the map contracts and fixes the section supplied
395 by the cover, Banach’s theorem gives uniqueness of that same fixed point.
396 The contraction rate is assumed, and the equality between the glued section
397 and the fixed point is part of E89. A coherent order parameter supplies the
398 supercritical inequality required by the stated contraction bound; it does not
399 construct the map or prove that cortical dynamics implements it.

400 The theorem `chain` concludes `UnifiedSelf` from these hypotheses. A
401 common toy witness satisfies all eight, so their conjunction is not empty.
402 The project declares no additional Lean axioms beyond the foundational
403 axioms used by Lean’s libraries; physical content resides in hypotheses and
404 model fields. This checks the conditional mathematical composition, not the
405 empirical truth of its premises. Detailed constructions, scope statements and
406 counterexamples are retained in the supplement and source files.

407 6 What the construction commits to about 408 consciousness

409 The composition terminates in a mathematical object, and we take a posi-
410 tion on what that object is. The commitment has two clauses. *Unity*: the
411 unique global section glued from compatible local states is the formal corre-
412 late of the unity of a conscious episode—the fact that a subject undergoes
413 one experience rather than a collection of co-occurring local ones. *Selfhood*:
414 the fixed point of the self-prediction map on that same section is the formal
415 correlate of a minimal self, a state that is what it takes itself to be. Neither
416 clause is a theorem, and no simulation here supports either. We argue for
417 them, then say what would count against them.

418 Three features of the construction motivate the identification, and each
419 distinguishes it from a bare synchrony measure.

420 First, compatibility rather than agreement. Oscillator phases can agree
421 everywhere while the local probabilistic descriptions they carry remain in-
422 compatible on their overlaps, and Section 5.2 exhibits that configuration
423 inside the formalization, not only as an illustration. If the unity of experi-
424 ence is a matter of local contents being about one world rather than merely
425 co-timed, then compatibility, not coherence, is the property to look for—and
426 coherence enters as the physical condition under which compatibility can be
427 maintained, which is what E78 asserts.

428 Second, uniqueness follows from the construction rather than from a sep-

429 arate selection principle. Accounts of conscious unity need something that
 430 picks one experience from among the candidates a system could support;
 431 integrated information theory postulates exclusion for this purpose (Tononi
 432 et al., 2016). Here uniqueness arrives twice as a consequence: given a cover
 433 whose local sections agree, the sheaf condition gives exactly one global sec-
 434 tion, and given a contracting self-prediction map, Banach’s theorem gives
 435 exactly one fixed point. The scope of that claim needs stating exactly, be-
 436 cause the stronger reading is available and is not ours. Uniqueness is relative
 437 to a cover and a map, and choosing them—which regions, which overlaps,
 438 which encoding region, and so where the subject ends—is not supplied by ei-
 439 ther theorem. It is what E78 and E89 carry. What the construction removes
 440 is the need for an additional principle to select among the experiences a fixed
 441 cover and map would otherwise admit; it does not remove the need to say
 442 which cover and map a given subject has. That is a structural argument,
 443 not evidence.

444 Third, the reflexive clause makes the self a condition rather than an addi-
 445 tion. The predictive map restricts a global state to a region encoding it and
 446 reads that encoding back out, so a fixed point is a state whose self-model is
 447 accurate about the state it models. That condition is available only because
 448 the state modelled is a single glued object: a system whose local descrip-
 449 tions do not cohere has no such state for a self-model to be about. Unity
 450 and selfhood are therefore one construction read at two levels, which is why
 451 `UnifiedSelf` requires the fixed point to be the glued section itself rather

452 than an unrelated section.

453 The costs belong with the argument. The commitment does not address
454 why any physical condition is accompanied by experience; it is a proposal
455 about which physical state is the conscious one, not about why states of that
456 kind feel like anything. It is therefore open to the objection that a system
457 could satisfy every condition—compatible local states, a supercritical field,
458 an accurate self-model—while nothing is experienced. We have no reply to
459 that objection and claim none. The identification is also multiply realizable
460 as stated: nothing in the sheaf or fixed-point conditions is specific to cortex
461 or to electromagnetic coupling, so the substrate claim is carried entirely by
462 E56 and E67 and is empirical. No conclusion about artificial systems follows
463 from the formal results.

464 What could count against the commitment is dissociation. It predicts that
465 the two clauses and the presence of a unified conscious episode go together,
466 so a state in which local cortical descriptions are demonstrably compatible
467 and the field supercritical, while the subject reports no unified experience,
468 would tell against it; so would unified reported experience where compati-
469 bility demonstrably fails. Anesthetic emergence, dreaming and dissociative
470 states are the places to look, because they separate coherence from report
471 in ways ordinary waking does not. Such tests need an operational measure
472 of overlap compatibility on neural data, which this article does not supply;
473 constructing one is the main obstacle to making the commitment empirically
474 live rather than merely stated.

475 The physical tests of Sections 3 and 4 are independent of all of this. They
476 concern coupling, coherence and recovery, and can fail while the identification
477 stands, or succeed while it is wrong.

478 **7 Discussion**

479 The framework connects thermodynamic information constraints, collective
480 phase organization and compatible local representations, conditionally and
481 with its assumptions named. Its biological content is a coupling-gated re-
482 covery hypothesis with defined observables, a stated protocol, and failure
483 conditions declared in advance. Its formal content is a composition whose
484 eight gaps are exhibited rather than argued past, and which can be inhabited
485 by a single model. Its interpretive content is the identification of Section 6,
486 offered as a commitment with reasons.

487 The simulations are useful because they restrict the hypothesis rather
488 than support it. Finite ramps smear the stationary onset to the point where
489 a fitted onset exponent chiefly reports the ramp; an unpinned collective
490 phase makes independence diagnostics ambiguous in exactly the supercritical
491 regime the model cares about; and squared-drift plasticity maintains coher-
492 ence while degrading structural alignment. The second and third of these
493 are results about method and about a common intuition, and they transfer
494 beyond this framework.

495 The proposal differs in emphasis from accounts based on integrated in-

496 formation (Tononi et al., 2016), global workspace dynamics (Baars, 1997;
 497 Dehaene et al., 2014), or predictive processing (Friston, 2010). It specifies a
 498 candidate interaction medium and a mathematical compatibility condition,
 499 and, relative to a given cover and self-prediction map, it derives the unique-
 500 ness those accounts postulate. It does not derive the exclusion principle of
 501 integrated information, identify every broadcast with a field transition, or
 502 derive a generative model from a dissipation bound. Continuous fields can
 503 carry incompatible local descriptions and discrete models can admit sheaf
 504 constructions, so continuity alone does not settle the combination problem;
 505 compatibility is what this account offers in its place.

506 Two formal results bear on the substrate question, and both admit a
 507 stronger reading than they support. At matched coupling resource, a fixed
 508 wiring support that excludes a better-correlated pair can be suboptimal—a
 509 statement about constrained optimization, not about biology. On an atom-
 510 less substrate, a kernel supported on finitely many points contributes zero to
 511 the chosen continuum coupling integral. That is a property of the measure
 512 and the observable, and it cuts against this framework’s own coarse-graining
 513 step as much as against anything else. It does not imply that digital hard-
 514 ware cannot approximate a field or support consciousness, and no biological-
 515 versus-artificial conclusion follows.

516 The empirical priorities follow directly from Section 3.2: estimate effec-
 517 tive coupling and diffusion in consistent units, test the proposed relation
 518 between coupling and extracellular geometry, and measure recovery against

519 an independently defined behavioral endpoint under matched model compar-
520 ison. A favorable concentration–coherence relation would support a statis-
521 tical description, not identify its physical cause; an unfavorable result in a
522 regime satisfying the stated assumptions would constrain that description.
523 Successful synchronization simulations cannot substitute for measurements
524 distinguishing ephaptic interactions from other coupling mechanisms.

525 The article therefore asks to be judged in three separable parts: a pre-
526 diction that can be tested, a composition that can be checked, and an iden-
527 tification that can be argued about. Keeping them separate is what makes
528 the whole assessable.

529 **Data and code availability**

530 The Lean development, simulation scripts, tests and saved numerical sum-
531 maries are maintained in the manuscript source repository. Publication sim-
532 ulation values are regenerated from saved summaries without rerunning pro-
533 duction sweeps. The exploratory EEG analysis uses OpenNeuro dataset
534 ds005620 (Bajwa et al., 2025); preprocessing and selection are described
535 in Supplemental Material. The source repository contains the reproducible
536 analysis and does not redistribute the raw recordings.

537 Use of AI tools

538 AI tools assisted simulation-code development and manuscript revision. Ma-
539 chine checking establishes the stated Lean results under their hypotheses;
540 it does not validate biological interpretations or replace scientific review of
541 generated code and prose.

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